

Hardware Component Troubleshooting & Diagnostics Report

Detailed step-by-step procedures for RAM reseating, GPU diagnosis, storage simulation, and random restart analysis.

Question 1

Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Step-by-Step RAM Reseating Procedure & Observations:

- **Step 1 (Safety Preparation):** Shut down the system completely, unplug the AC power cord, disconnect all peripherals, and press the power button for 10 seconds to drain residual motherboard capacitor power.
- **Step 2 (Chassis Access):** Opened the computer side panel or laptop back cover by removing securing screws, ensuring an anti-static wrist strap was worn to prevent electrostatic discharge (ESD).
- **Step 3 (Locating DIMM Slots):** Located the memory slots on the motherboard, noting the dual-channel configuration and identifying the target RAM stick.
- **Step 4 (Unlatching & Removal):** Pressed down firmly on the plastic retention clips located at both ends of the DIMM slot until they clicked outward, then gently pulled the RAM module straight up out of the slot.
- **Step 5 (Re-insertion & Verification):** Aligned the notch on the RAM module with the key inside the slot, inserted it vertically with even downward pressure until the side retention clips snapped back into place automatically.
- **Observed Issues / Notes:** Minor dust accumulation was observed inside the DIMM slots (cleaned using compressed air), and stiff retention latches required careful manual guidance to avoid bending components.

Question 2

Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

GPU Troubleshooting & Diagnostic Checklist:

- **Step 1 (Symptom Identification):** Notice initial gaming issues such as graphical artifacts (color glitches, pink squares), sudden game crashes to desktop, screen tearing, or complete lack of video output (no signal on monitor).
- **Step 2 (Physical & Cable Connection Check):** Verify that the HDMI or DisplayPort cables are securely connected directly into the discrete GPU ports rather than the motherboard, and check that auxiliary PCIe power cables from the PSU are firmly plugged in.
- **Step 3 (PCIe Slot & Seating Inspection):** Power down, open the case, unlatch the PCIe slot retention clip, remove the graphics card, inspect for dust or burnt contacts, and firmly re-insert it until secure.
- **Step 4 (Driver Clean and Reinstall):** Boot into Windows Safe Mode, use Display Driver Uninstaller (DDU) to completely wipe corrupted graphics drivers, restart, and install the latest stable WHQL driver package from NVIDIA or AMD.
- **Step 5 (Confirming the Fix):** Run a stress-testing benchmark utility (such as FurMark or 3DMark) to monitor GPU temperatures, clock stability, and fan speeds under heavy load, verifying zero crashes or visual artifacts.

Question 3

Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Storage Simulation & Diagnostic Indicators:

- **Before Reconnecting (Unplugged State):** When booting the system with the storage drive disconnected, the motherboard fails to detect any bootable media. The screen displays error messages such as *"Reboot and Select proper Boot device or Insert Boot Media in selected Boot device and press a key"* or the system goes straight into the UEFI/BIOS setup utility where the storage slot/SATA port shows as empty or "Not Detected".
- **After Reconnecting (Plugged State):** Once the SATA data/power cables or NVMe M.2 screw and slot connection are properly re-secured, entering the BIOS reveals the drive listed correctly with its exact model number and capacity. Upon saving and exiting, the system successfully loads the operating system kernel without error.

Question 4

Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Component Selection, Reasoning & Diagnostic Steps:

- **Component Checked First: RAM (Random Access Memory).**

- **Reasoning:** Random reboots or sudden Blue Screens of Death (BSOD) with memory-related error codes (e.g., MEMORY_MANAGEMENT) are most frequently caused by unstable memory timings, faulty DIMM modules, or loose seating. RAM is also the quickest and easiest component to test and reseat compared to a GPU or storage drive.

- **Specific Diagnostic Steps:**

1. Power down the PC, open the chassis, and reseat all RAM sticks into alternative or single-channel slots.
2. If multiple sticks are installed, test them one by one to isolate a potential bad module.
3. Boot into a hardware diagnostic utility like **Windows Memory Diagnostic** or run a bootable **MemTest86** USB drive through multiple passes to check for memory bit errors.
4. Reset BIOS memory profiles (disable unstable XMP/EXPO overlocks) to verify if factory stock speeds resolve the random reboot behavior.